

Concurrent Training Intensities: A Practical Approach for Program Design

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ABSTRACT

Sports performance is influenced by the interaction of several physical variables. For this reason, most sports need both strength and endurance capacities to maximize overall performance. Therefore, a combination of resistance and aerobic training, usually called concurrent training (CT), has been used recently as a way of simultaneously improving strength and aerobic performances according to the needs of a specific sport. This combination can be challenging and can influence training adaptations, being a problematic issue for coaches. The main objective is to provide coaches with a practical proposal for CT to improve athletes' performance in different sports.

INTRODUCTION

The performance of physical activities aiming to develop both aerobic capacity and strength within the same training session, or in different sessions, is usually termed as

concurrent training (CT) and has been a research target in recent years (21). Early studies have pointed out that CT might compromise aerobic and/or strength gains (6,9,19). However, recent findings have suggested that, on the contrary, CT may actually enhance individual performance in aerobic and strength parameters (5,34,35). The discrepancies between these studies may be justified by their design and/or training protocols (14). Therefore, it is important to combine the several variables in a proper way to obtain more conclusive results.

Interference between strength and aerobic training can be caused by several factors associated with the training program, such as the volume, intensity, and/or training load distribution (8). Regarding the volume, it was found that strength gains were compromised by high weekly training volume, considering frequency and/or duration of exercise (20). Unfortunately, little evidence exists regarding the manipulation of the intensities of aerobic and/or strength training performance, and the evidence that does exist is recent (36,37). Consequently, this could be a major issue when programming both aerobic and

resistance training in sports where these 2 variables are crucial.

CT intensity has been reported as a variable that could influence performance in both aerobic and strength parameters, particularly when aerobic and strength training are combined (13,27). For example, previous studies have argued that low-volume, high-intensity strength training alone (e.g., maximal strength training or plyometric/explosive resistance training) could induce greater improvements in aerobic and strength performance than moderate-intensity training (31).

Strength and conditioning professionals have been advised to prescribe programs that include both strength and aerobic training concurrently to obtain better results with more efficiency and quickness (21). For this reason, coaches and professionals should know how to program a specific CT regarding volume, intensity, duration, periodization models, to conjugate the loads, and to obtain increased performances. Thus, the main objective of this article is to

KEY WORDS:

aerobic; resistance; detraining; programming

provide the knowledge and recommendations to enable coaches to efficiently design a CT training regime that will improve sports performance.

INTENSITIES DURING CONCURRENT TRAINING

Several studies have shown a performance enhancement after 8 weeks of CT (40,42). Moreover, this performance was accomplished by strength and aerobic gains after the training period, and this suggests the beneficial effect of CT after a short period of implementation. This reveals that CT can be used for strength and aerobic development. However, some concerns should exist regarding the combination of resistance and aerobic training loads. For instance, Souza et al. (12) have concluded that maximal strength gains and muscle endurance may be compromised when training occurs in combination with high-intensity aerobic exercises (close to maximal oxygen uptake). Similar findings were observed by Chtara et al. (9). These authors found a reduction in strength and muscular power output after 12 weeks of CT comprising high-intensity aerobic exercises. Both authors suggest that the decrease in performance was caused by the fatigue generated during aerobic training, which compromised either the resistance training or the muscle adaptations (9,12).

The interference effect was investigated in depth by Kraemer et al. (23), who examined the morphological adaptations of muscle fiber during CT over 12 weeks (4 times per week) in physically active men. They verified that there were increases in type I, IIA, and IIC fibers in the group that only did resistance training, increased IIA fibers in those who did resistance and aerobic training simultaneously, and a decrease in type I and IIC fibers in those who only performed aerobic training. These results indicated that there might be a decrease in adaptation to resistance training when it is combined with an aerobic training regimen. It seems that resistance training enhances aerobic training, but not vice versa. This was confirmed by others who suggested that CT was more effective in

improving aerobic performance than aerobic training alone (16,23).

In the available literature, we also find that there is no interference of one ability over the other when training concurrently (2,4). Alves et al. (2) compared the effects of strength training alone, intrasession of combined strength and aerobic training, and intersession of combined strength and aerobic training. The results showed that both groups that performed strength and aerobic training concurrently obtained larger gains in explosive strength and aerobic capacity when compared with the group that performed strength training only. In the study of Balabinis et al. (4), the group that trained concurrently obtained similar gains in strength and muscular power when compared with the group that only performed resistance training. This inconsistency regarding the CT interference effect is now understood as the result of several programming factors, where the intensity of aerobic and/or resistance training should be considered the main issue.

Intensity is usually seen as a major influence on training programs and adaptations (38). The change in training intensity could affect the magnitude of molecular signaling and protein synthesis (13), which will therefore influence the degree of interference between exercise modes and can also vary depending on programming variables (11,13). It is only recently that research has focused on this issue in CT programming, and only a few studies have compared combinations of different intensities in resistance or aerobic training seeking to find the combinations that lead to the greatest enhancements (27,36,37,40). Most studies have tried to compare different training load distributions and different methods of training, making it hard to understand the reasons for the training adaptations.

To the best of our knowledge, Sousa et al. (36,37) were the first to investigate the effects of using different resistance training loads or different aerobic intensities and the effects of a detraining

period on strength and aerobic variables, respectively. Regardless of the training intensity (low, moderate, or high), resistance training combined with low-intensity aerobic training resulted in beneficial effects for both strength and aerobic development (36). However, the authors found that choosing higher loads during resistance training can also lead to explosive increments during vertical jump and short sprint efforts (36). When combining resistance training with high-, moderate-, or low-intensity aerobic training, it was the lowest intensity that resulted in the highest gains in maximal strength, with similar gains in cardiorespiratory fitness (37). Moreover, combining moderate-to-high resistance loads with low-intensity aerobic training revealed smaller losses after 4 weeks of training cessation (37).

PROGRAM DESIGN: AN EXAMPLE

Most individual and team sports require multiple physical capacities for optimal performance, such as muscle strength, speed, power, and cardiorespiratory fitness (41). For some sports, success only arrives when a good performance is achieved in all capacities (41). In fact, athletes must be physically prepared for repeated sprints (10), jumps, changes of direction (24), throws, and shots (15,25). Most athletes need to develop strength to apply their abilities, as well as cardiorespiratory fitness to recover and/or maintain high levels of performance throughout the game, event, or season (24).

Several studies have shown that a CT program design could be safely implemented to improve a wide variety of athletic performance variables, such as strength and aerobic fitness (3,26). For example, a 7-week CT program was effective in improving vertical jump performance in basketball players, compared with isolated resistance training (4). Other studies conducted with elite soccer players during the preseason period showed that CT improved explosive strength (18). Another study conducted among highly trained individuals reported that CT induced strength gains after 6

weeks of intervention (27). Finally, 8 weeks of CT effectively enhanced explosive performances and aerobic endurance in professional soccer players (42).

Regarding the intensities used during CT, Sousa et al. (36) reported that 8 weeks of resistance training programs combining different aerobic and resistance training loads improved strength and aerobic capacities. Nevertheless, it was suggested that resistance training with loads higher than 55% of 1 repetition maximum (1RM) maximized the explosive efforts gains, with higher gains at higher loads (e.g., countermovement jumps [CMJ] and short sprints). Knowing this, we compared the use of the same resistance training loads combined with different aerobic training intensities (37). It was found that they improved aerobic capacity similarly, but higher gains in maximal strength were found in the low-intensity aerobic training group, compared with the moderate- and high-intensity groups (37).

Based on recent findings from our laboratory and others, we present a practical example for 8 weeks of implementation of CT, performed twice a week (Table 1). Therefore, we provided a practical application to enhance lower extremity strength, muscular power, and aerobic performance. CT can be used in individual or team sports in which those physical abilities are required, such as jumps, repeated sprints, or maximum lower-limb strength. Our research design respected 3 main general principles of progression: progressive overload, variation, and specificity as recommended by Kramer and Ratamess (22). The first weeks were designed around the initial adaptation to training and, at the same time, the enhancement of explosive strength gains, focusing on lower resistance training loads and lower aerobic training volume. For example, data from previous research (1) suggest that adaptations within the fast fiber production (IIb to IIa) occur during the early phases of training although there were no changes in slow to fast fibers.

Table 1
Example of an 8-week concurrent training program to improve lower-body strength and cardiorespiratory performance

Exercise	Sessions															
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Full squat (% 1RM: S × R)	70.3 × 6	70.3 × 6	70.3 × 8	70.3 × 8	75.3 × 6	75.3 × 8	75.3 × 8	75.3 × 8	80.3 × 5	80.3 × 5	80.3 × 5	80.3 × 5	85.3 × 5	85.3 × 5	85.3 × 5	85.3 × 5
CMJ (S × R)	2 × 4	2 × 4	2 × 5	2 × 5	2 × 5	2 × 5	2 × 5	2 × 5	2 × 5	3 × 5	3 × 5	3 × 5	3 × 5	3 × 5	3 × 5	3 × 5
Sprint (S × D)	2 × 30 m	2 × 30 m	2 × 30 m	2 × 30 m	3 × 30 m	3 × 30 m	3 × 30 m	3 × 30 m	3 × 20 m	3 × 20 m	3 × 20 m	4 × 20 m	4 × 20 m	4 × 20 m	3 × 20 m	3 × 20 m
20-m Shuttle run (total duration and intensity as % MAS)	16 min (80%)	16 min (80%)	16 min (80%)	16 min (80%)	16 min (80%)	16 min (80%)	20 min (80%)	20 min (80%)	20 min (80%)	20 min (80%)	20 min (80%)	20 min (80%)	20 min (80%)	20 min (80%)	20 min (80%)	20 min (80%)
1RM = 1-repetition maximum; CMJ = countermovement jump; S × R = sets × repetitions; S × D = sets × distance; %MAS = percentage of the maximal speed reached for each participant during the 20-m multistage shuttle run test.																

Considering that training programs should be specific and according to training objectives (32), we included full squats (FS), CMJ, and sprints to contribute to anaerobic muscle adaptation to training. The rest period between each set and exercise should be 2–3 minutes (25,36). Participants should be instructed to perform all exercises at maximal intended velocity to obtain the highest possible gains (19). The loads used by each participant in the FS were assigned according to 1RM obtained in the initial isoinertial squat strength assessment. Thus, the relative intensity of the FS exercise progressively increased from 70 to 85% 1RM. Because strength was expected to increase with training, an intermediate strength assessment could be conducted after 4 weeks of training to perform the necessary load adjustments for each athlete. Between the resistance and the aerobic training, each athlete should rest for 15–20 minutes so that the required

intensities can be performed properly (23). This training component was designed to be simple to evaluate and control in a real context, consisting of 16–20 minutes of 20-m shuttle run exercise at 80% of maximal individual speed reached during a previous 20-m multistage shuttle run test.

In our program, we used a low-repetition model to allow the athletes to reproduce all repetitions at maximal intended velocity. If the aim is to increase the load, coaches are recommended to change the external load or increase the number of sets but not the number of repetitions (25). On this subject, Campos et al. (7) compared the effects of 3 different resistance-training programs on adaptations within the vastus lateralis muscle over 8 weeks. All training regimens caused similar changes in fibers, from IIB to IIA conversions, but curiously, they found that maximal dynamic strength improved the most with low-repetition

training (compared with medium and high repetitions), raising the question of whether more repetitions (more exercises) are not necessarily good. This way, considering that the suggested CT program requires 72 hours of recovery between each session (22), it is expected that athletes can perform the specific training for their sport in a nonfatigued state and thus maximize their technical ability.

We chose to provide an example of 8 weeks because although the nervous system plays a significant role in the early phases of adaptation to training (33), within a period of 4–8 weeks of training, muscle hypertrophy becomes evident (23,28). Moreover, most of the training regimens used cycles of 4–8 weeks of preparation, not only in competitive training but also during strength training programming. Many investigations have used an eight-week period as a baseline. For example, Botonis et al. (5)

Table 2
Recommendations for combining intensities during concurrent training

	Recommendations
General	A minimum of 6-week periods for positive effects;
	At least 2 times a week in a concurrent design (resistance and aerobic);
	Training should comprise specific exercises according to each sport
	Two weeks of detraining is allowed without performance losses (i.e., injury, recovery)
Resistance training component	High velocity movements should be used;
	Exercises with external loads (e.g., squat or bench press) combined with ballistic movements (e.g., ball throwing, jumps);
	Ballistic exercises should be included in the beginning;
	Few repetitions per set (<8) and large rest intervals (>2 min);
	Progression should increase the number of sets instead of repetitions;
	External loads should be moderate-high (>55% 1RM).
Aerobic training component	Aerobic training should follow the resistance training;
	Low intensities (<LT or <75% maximal aerobic capacity) should be used;
	High intensities (> $\dot{V}O_{2max}$) should be performed for cardiorespiratory gains;
	Polarized model for training should be followed during a season (65% below ventilatory threshold and 10% above respiratory compensation threshold)
LT = lactate threshold; RM = repetition maximum; $\dot{V}O_{2max}$ = maximal oxygen uptake.	

recently observed that CT with high-intensity aerobic training during pre-competition season seems to be an effective regimen to enhance swimming performance. They used a training program of 8 weeks, the time that a precompetition preparation could last for water-polo season in real context. Recently, Rivi re et al. (30) studied the use of different resistance training with elastic bands and found that 6 weeks of CT could lead to improved upper-body strength, velocity, and power in elite youth rugby players. Although our example comprised 8 weeks, we think that this can be expanded and replicated to a longer period.

Each training phase usually lasts 2–10 weeks, and the complete training cycle ranges from 8 to 35 weeks (17). If we consider periods of prolonged competition (e.g., 26 consecutive weeks), it would require intensity manipulation on a weekly or micro-cycle basis. Periodization is the systematic variation of volume and intensity (29), although it is problematic to discuss those things separately from one another. It has been demonstrated that the fluctuation of workload increases can stimulate performance gains (39). In fact, the periodized variation with the specific sequencing of the volume and intensity offers an optimized method of improvement (39). In our opinion, knowing this and the effects of CT after a detraining period, a minimum of 8 weeks should be spent on CT and a maximum of 2 weeks should be spent on unloading, to avoid any loss of previous gains in strength and aerobic variables. This detraining period could also allow a maximization of previous gains by a supercompensation effect (43). It is also important to be aware that some of the latest research on intensity during CT focuses on the distribution of the load throughout a season, and a polarized model is suggested (40).

This CT training could also be implemented for team sports, in combination with refining work and technical and tactical conditioning. It could be applied during a critical time of the season for players who often compete 2 or 3 times a week in national and/or international competitions. However, in these cases, the physical and psychological recovery time should be considered. Furthermore, in Table 2 we provide some recommendations for helping coaches to design their own CT program.

SUMMARY

Studies of CT have shown that it has beneficial effects on athletes' performance, once properly combined. Thus, the intensity of resistance and/or aerobic training seems to play an essential role in greater gains. The few extant studies gathered showed improved strength and cardiorespiratory performance, regardless of the different intensities used in aerobic and/or resistance training during CT. However, there is a tendency for higher neuromuscular gains when higher-resistance training intensities are combined with low-intensity aerobic training. Also, higher aerobic adaptations were found when lower aerobic intensities were used. Knowing this and the fact that detraining periods longer than 2 weeks can compromise previous gains, coaches can design a CT program that aims to improve strength and cardiorespiratory fitness and optimize athletic performance.

Conflicts of Interest and Source of Funding: The authors report no conflicts of interest and no source of funding.

ACKNOWLEDGMENT

This project was supported by the national funds through FCT–Portuguese Foundation for Science and Technology (UID/DTP/04045/2019)–and the European Fund for Regional Development (FEDER) allocated by the European Union through the COMPETE 2020 Programme (POCI-01-0145-

FEDER-006969)–competitiveness and internationalization (POCI). The authors disclose funding received for this work from any of the following organizations: National Institutes of Health (NIH); Wellcome Trust; Howard Hughes Medical Institute (HHMI); and other(s).



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